



**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE & TECHNOLOGY**

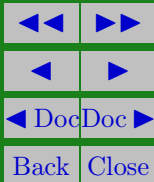
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ICS 2302 SOFTWARE ENGINEERING 1





This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 9

Quality Management

Learning Outcomes

A study of this lesson should enable students to:

- Describe Quality Management



ICS 2302 Software Engineering 1

Achieving a high level of product or service quality is the objective of most organizations. Quality management involves defining procedures and standards which should be used during software development and checking that these are followed by all engineers. Quality management provides an independent check on the software development process. The deliverables from the software process are input to the quality management process and are checked to ensure that they are consistent with organizational standards and goals. Software quality management can be structured into three principal activities:-

1. Quality assurance
2. Quality planning
3. Quality control



Quality Assurance and Standards

Quality assurance (QA) activities define a framework for achieving software quality. The QA process involves defining or selecting standards that should be applied to the software development process or software product. These standards may be embedded in procedures or processes which are applied during development. There are two types of standards that may be established as part of the quality assurance process:

1. Product standards: These are standards that apply to the software product being developed. They include document standards such as the structure of the requirements document which should be produced, documentation standards such as a standard header for an object class definition and coding standards which define how a programming



language should be used.

2. Process Standards: These are standards that define the process which should be followed during software development. They may include definitions of specification, design and validation process and a description of the documents which must be generated in the course of these processes.

There are a number of reasons why software standards are important:

- They provide an encapsulation of best practice.
- They provide a framework around which the quality assurance process may be implemented. Given that standards encapsulate best practice, quality control simply involves ensuring that standards have been properly followed.



Quality Planning

Quality planning should begin at an early stage in the software process. A quality plan should set out the desired product qualities. It should define how these are to be assessed. It therefore defines what high quality software actually means. Without such a definition, different engineers may work in an opposing way so that different product attributes are optimized. The result of quality planning process is a project quality plan. The outline structure for a quality plan should include:

1. Product Introduction- A description of the product, its intended market and the quality expectations for the product.
2. Product Plans - The critical release dates and responsibilities



ICS 2302 Software Engineering 1

ities for the product along with plans for distribution and product servicing.

3. Process Description – The development and service processes which should be used for product development and management.
4. Quality Goals - The quality goals and plans for the product, including an identification and justification of critical product quality attributes.
5. Risk and Risk Management- The key risks which might affect product quality and the actions to address these risks.



Quality Control

Quality control involves overseeing the software development process to ensure that quality assurance procedures and standards are being followed. There are two complementary approaches to quality control:

1. Automated Software Assessment - Where the software and the documents which are produced are processed by some program and compared to the standards which apply to that particular development project.
2. Quality Reviews - Where the software, its documentation and the process used to produce that software are reviewed by a group of people. They are responsible for checking that the project standards have been followed and that



software and documents conform to these standards. Deviations from the standards are noted and brought to the attention of project management.

Software Measurement and Metrics

Software measurement is concerned with deriving a numeric value for some attribute of a software product or a software process. By comparing these values to each other and to standards which apply across an organization, it is possible to draw conclusions about the quality of software or software processes.

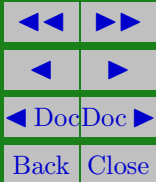
A software metric is any type of measurement which relates to a software system, process or related documentation. Metric may be either control metrics or predictor metrics. Control metrics are usually associated with software processes. Predic-



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tor metrics are associated with software products.





Revision Questions

Example . Which of the following is the most efficient means of product quality inspection?

Acceptance sampling

100% inspection

Variable sampling

a and c

Solution: Variable sampling



EXERCISE 1.  The primary components of the quality management function are ?

quality planning

quality control and assurance

quantitative measurement



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quantitative assessment of the project
a and b

References and Additional Reading Materials

1. Hans van Vliet (2005). Software Engineering: Principles and Practice (2nd ed.). John Wiley & Sons, Inc. ISBN: 0-471-97508-7
2. Ian Sommerville, Pete Sawyer(2005). Requirements Engineering: A Good Practice Guide. ISBN: 0-471-97444-7. John Wiley & Sons, Inc
3. Gerald Kotonya, Ian Sommerville (1998). Requirements Engineering: Processes and Techniques. John Wiley & Sons, Inc. ISBN: 0-471-97208-8



Solutions to Exercises

Exercise 1.

quality planning

quality control and assurance

Exercise 1